Final Year Project · Planning Report: LLM Trading Agent Evaluation on A-shares

Project Goal

Build a public, continuously running evaluation and training ground for LLM trading agents. Users submit an agent configuration; it researches, decides and trades on contamination-resistant A-share data and returns a verifiable scorecard and decision record. Success means: a demonstrable end-to-end loop producing an equity curve and per-trade decision record; one discriminating result, where models score similarly on decision quality yet differ on decision flip rate; and no ranking where samples are too few, labelled "insufficient evidence". Out of scope: live trading and stock tips.

Significance

What. The facility gives agents rule-correct A-share matching and accounting, point-in-time masked data, and a logged decision record. Users submit a model, prompt, tool set and risk gate; the system reports where a configuration is unreliable.

Who. Primary users are individuals already using AI to support investment decisions, including researchers and small-institution practitioners, who cannot check whether a conclusion holds up. Secondary users are developers.

Why. An A-share AI trading comparison on Bilibili reached 95,000 views, and when HKU's AI-Trader added A-shares to its benchmark, none of six models beat the SSE-50 baseline. The slot is open: public arenas cover crypto and US equities, and TraderHarness stays local.

Problem Statement

An investor cannot judge whether an AI recommendation is reliable. Existing tools give conclusions without a traceable process, leaving three questions unanswered: which data produced it, whether a rerun later would change it, and whether following it would have made money.

Leaderboards rank on return alone, and return is the weakest signal. Across 56 models and 2,826 trades on TradeRank only 46.2% of model-seasons were profitable, and all six models in HKU's A-share benchmark underperformed. A strategy can post a flattering in-sample Sharpe and turn negative out of sample; without that split, much of the result is noise.

The protocol is untrustworthy: models may recognise the tested period and companies from training memory, and general backtesting frameworks ignore A-share rules, so T+1 settlement, price limits, call auction and dividend adjustments must be rebuilt. Nobody has made it public and reproducible. The project asks whether evaluation can move beyond who earned most.

Proposed Solutions

The proposal combines evaluation and training. Baseline capability (market data, stock analysis, reports, MCP and Skill integration) is the entry ticket; the arena core (matching engine, simulated accounts, decision logging, contamination resistance, multi-agent co-trading, metric set) must be right; the differentiating layer (health report, tool ablation, leaderboard exposure effect, synthetic markets) is what competitors cannot build.

#	Deliverable	Acceptance test
D1	A-share matching and accounting engine	T+1, limits, call auction, fees and dividends correct; reproducible
D2	Contamination-resistant data layer	Point-in-time masking at every exit; leak audit clean
D3	Decision ledger	Reasoning, evidence and tool calls logged per decision
D4	Metric set and health report	Return, risk and process metrics; no ranking below 20 trading days
D5	Multi-agent co-trading and tool ablation	Shared capital, isolated accounts; tool-removal metric delta
D6	Demo and open-source repository	Create agent, enter, view equity, export report

Proposed Timelines

Weeks count from the project start. Phase 1 (weeks 1-4): technology selection, data source trials, evaluation protocol (technical plan). Phase 2 (5-10): data layer, matching engine, simulated accounts (D1). Phase 3 (11-14): decision ledger, contamination resistance (D2, D3). Phase 4 (15-18): metric set, multi-agent co-trading, and a 3-model by 5-stock by 20-day validation experiment (D4, D5). Phase 5 (19-22): blind cohort experiment, demo sign-off, release (D6).

Three risks handled by design: leaderboard and health report scored separately; synthetic markets skipped; the A-share slot taken first. No investment advice.